Project Introduction

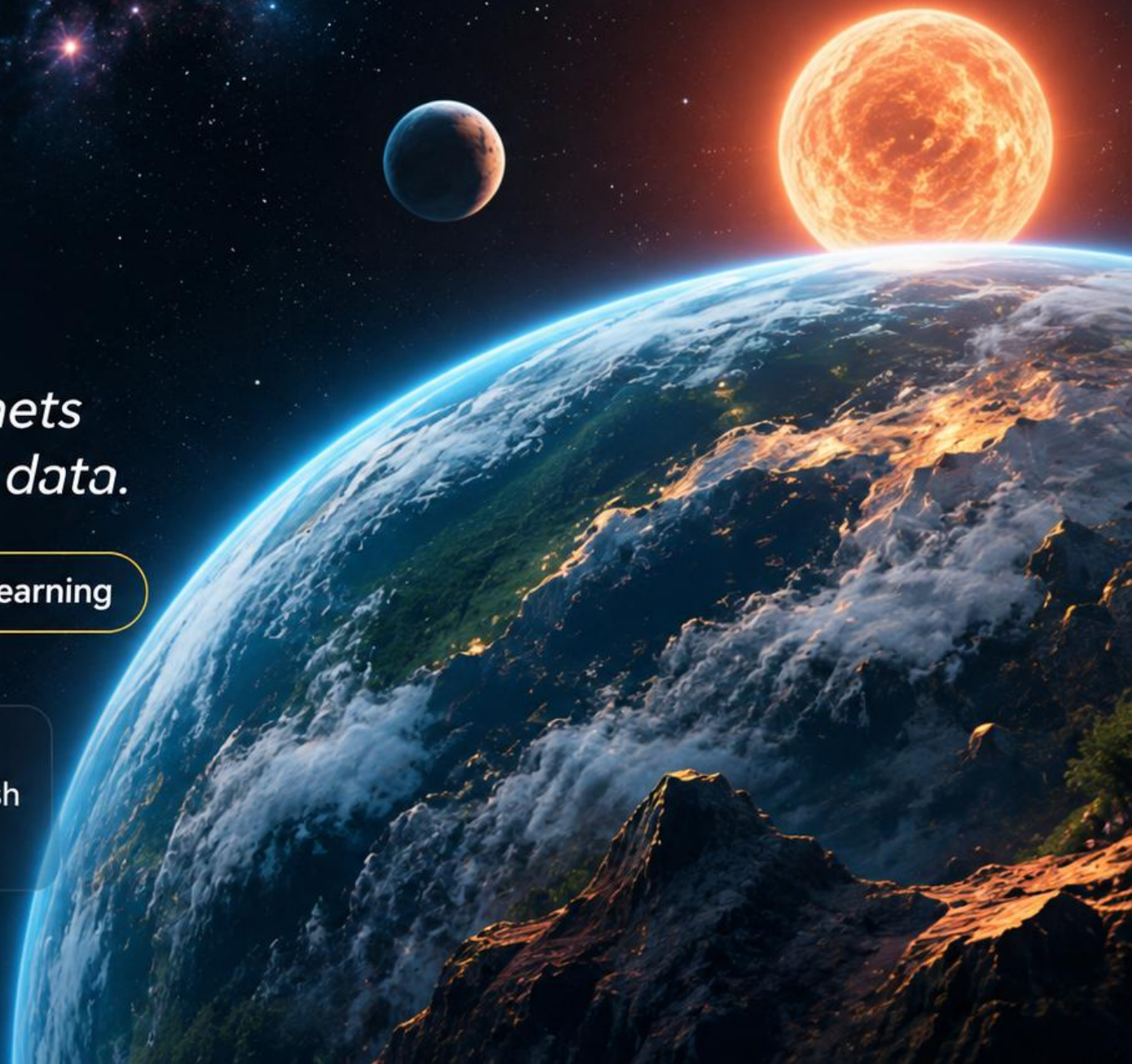
ExoVet

*Teaching a model to tell real planets from impostors in **NASA's Kepler** data.*

Astronomy

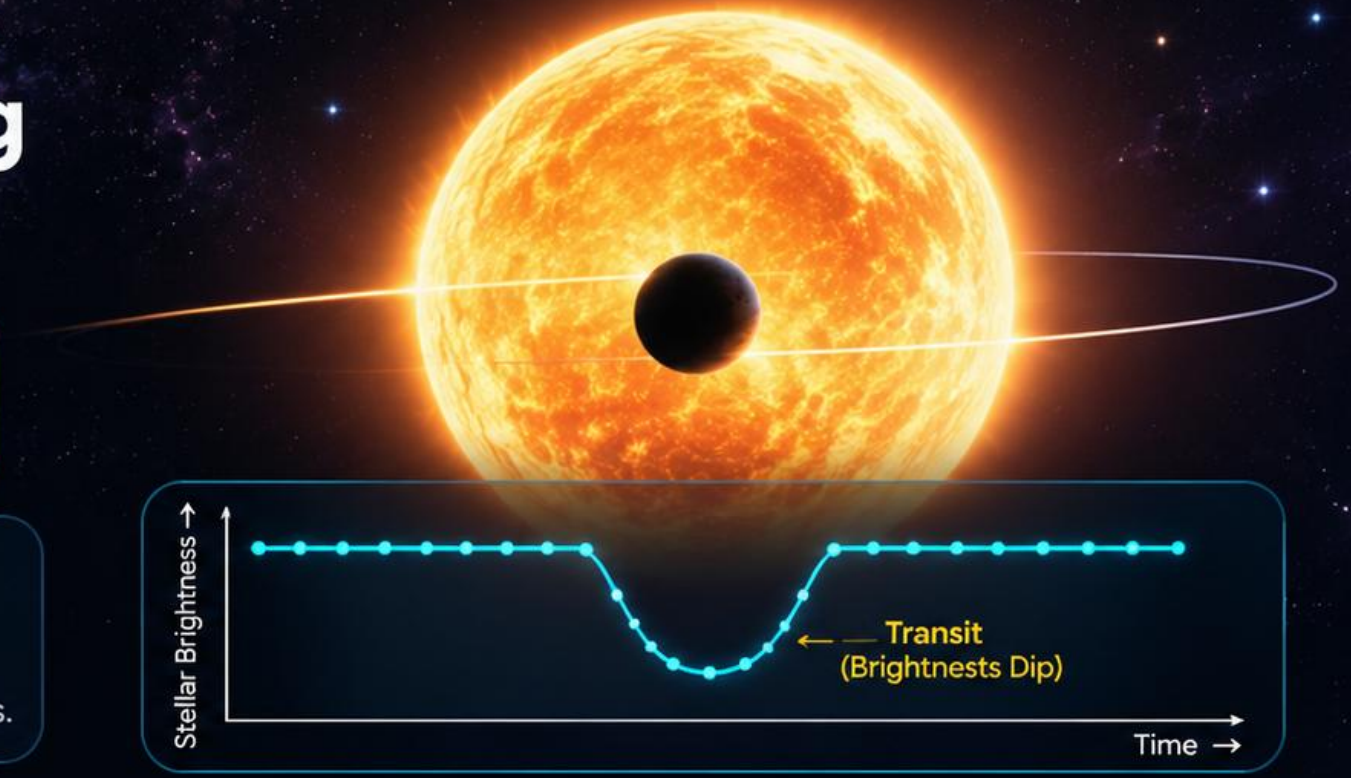
Space Science + Machine Learning

An AI-powered exoplanet vetting platform that analyzes **NASA's Kepler** observations to distinguish genuine planets from false-positive signals.



The Problem

Why is Exoplanet Vetting So Challenging?



What are Exoplanets?

Planets that orbit stars outside our solar system.

The Kepler Mission



NASA's Kepler Space Telescope monitored ~150,000 stars for four years, detecting tiny, repeated dips in starlight caused by orbiting planets.

Why Do *False Positives* Occur?



Eclipsing Binary Stars



Background Blended Stars



Instrumental Noise & Stellar Activity

Why is it Difficult?

- Many signals look similar to real planetary transits.
- Manual review is slow and doesn't scale.
- Future missions (TESS, PLATO, Roman) will generate even more candidates.
- Requires a systematic, automated approach to flag real planets efficiently.

The Solution

ExoVet: A Machine Learning Solution

A data-driven platform that uses machine learning to automatically vet exoplanet candidates from NASA's Kepler observations, identify real planets, and explain its decisions.



Learn Patterns from Data

Classify Candidates (Planet / Impostor)

Predict Probability of Being a Planet

Explain Key Factors

Kepler Dataset

NASA Kepler Objects of Interest (KOI)

~ 9,564 candidates

→

Machine Learning Models

- Logistic Regression
- Random Forest
- HistGradientBoosting

→

Prediction

- Confirmed Planet (Real)
- False Positive (Impostor)

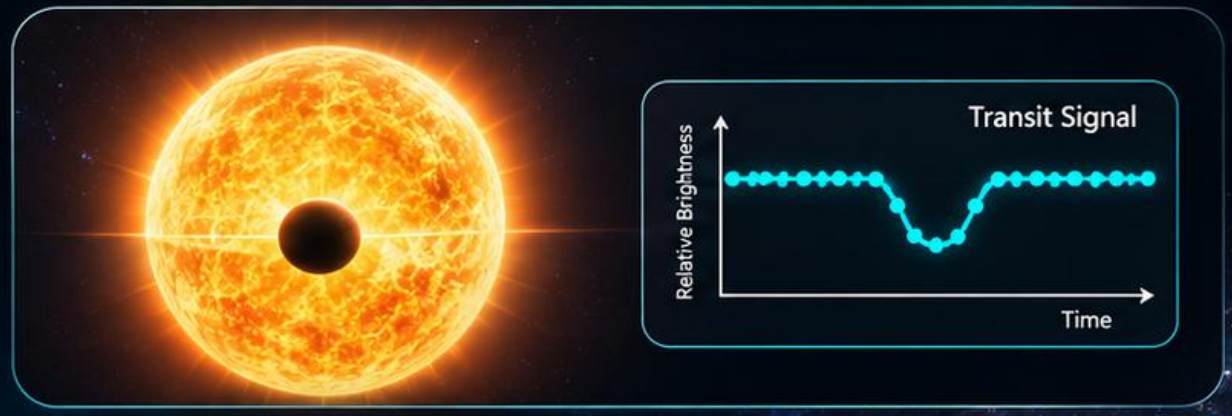
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Interpretation

- Planet Radius
- Transit Depth
- Signal-to-Noise
- Orbital Period
- ...

What ExoVet Does

- Uses Kepler catalog measurements (period, depth, duration, radius, SNR, etc.).
- Trains and compares multiple ML models.
- Predicts the probability of a candidate being a real planet.
- Explains which features drive the prediction.



The Technology

Technology Behind ExoVet

A modern full-stack and machine-learning pipeline to vet exoplanet candidates from NASA's Kepler data.

Python & Scikit-learn

Data processing, feature engineering, model training, and prediction.

Next.js & FastAPI

Modern web interface with a high-performance Python API for real-time inference.

Three.js

Immersive 3D solar system, planetary visualization, and interactive space experience.



Machine Learning Models

Random Forest

Robust classification with high generalization.

HistGradientBoosting

High-performance gradient boosting for accurate planet vetting.

NASA Exoplanet Archive

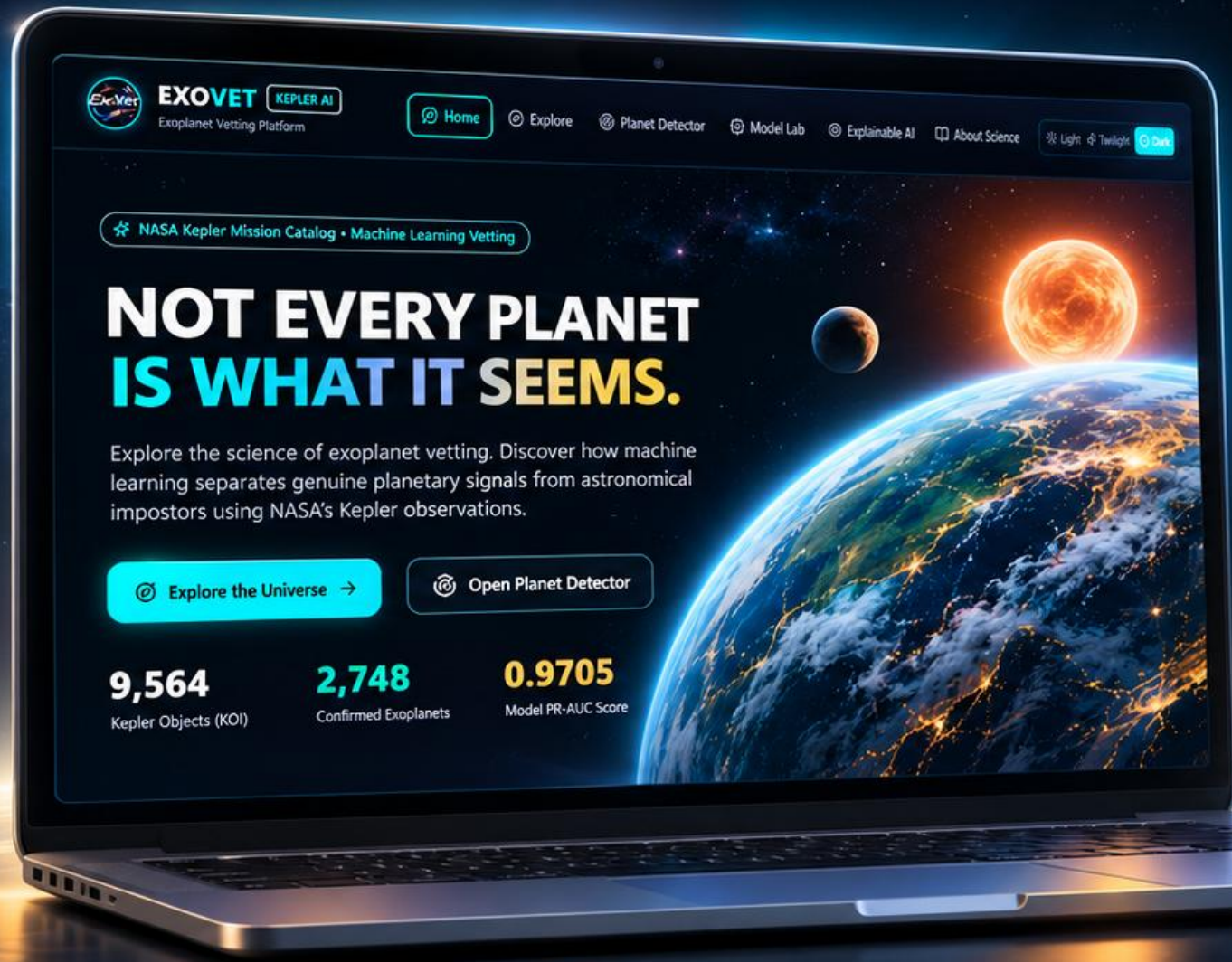
Kepler Objects of Interest (KOI) catalog with ~9,564 planetary candidates and their photometric measurements.

Kepler Data → ML Models → Web Platform → Planet Predictions & Insights

Introducing the Platform

Introducing the *ExoVet* Platform

A complete, interactive web platform to explore, analyze, and vet exoplanet candidates using real data and machine learning.



What You'll *Explore*

Home	Project overview & 3D solar system
Explore	Browse 9,564 Kepler candidates
Planet Detector	Get ML predictions & confidence
Model Lab	Compare models & view performance
Explainable AI	Understand key features & results
About Science	Learn the transit method & try an interactive simulator

Live Walkthrough →